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CSR-induced emittance growth in achromats: Linear formalism revisited

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ABSTRACT

We review the R-matrix formalism used to describe Coherent Synchrotron Radiation (CSR)-induced projected emittance growth in electron beam transport lines and establish the connection with a description in terms of the dispersion-invariant function.

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1. Introduction

Coherent Synchrotron Radiation (CSR) is well known to represent a fundamental limitation to the preservation of electron beam brightness during transport or magnetic compression [1]. As an electron bunch travels through a dipole, synchrotron radiation is emitted coherently at wavelength comparable to the bunch length, and unless suppressed by a narrow vacuum chamber, it may intoduce a significant modification of the beam particle energy depending on position along the bunch. The presence of dispersion correlates the energy change with a transverse offset in the bending plane (position, angle) resulting into emittance growth. Because of the correlated nature of the energy change, the effect, unlike the one stemming from incoherent synchrotron radiation, is in principle reversible. This principle has long been realized and often exploited in the design of beamlines [2–11]. The purpose of this paper is to review aspects of the 'R-matrix' [2] used to model the CSR-induced emittance growth, restating some general conditions for the attainment of minimum emittance growth, and clarifying its connection with the dispersion-invariant ('curly'-$\mathcal{H}$) formalism sometimes used to estimate the magnitude of the effect [8,9].

2. R-matrix formalism

2.1. Correlated emittance growth

Consider a beamline extending from $s=s_\mathrm{i}$ to $s=s_\mathrm{f}$. The transverse coordinates of a particle having longitudinal coordinate z_s and experiencing a CSR-induced relative-energy change $\delta_s(z_s)$ at location $s\in[s_\mathrm{i},s_\mathrm{f}]$ evolve according to

$$x(s_\mathrm{f})=R_{11}^{s_\mathrm{i}\to s_\mathrm{f}}x_\mathrm{i}+R_{12}^{s_\mathrm{i}\to s_\mathrm{f}}x_\mathrm{i}'+R_{16}^{s\to s_\mathrm{f}}\delta_s(z_s),$$
$$x'(s_\mathrm{f})=R_{21}^{s_\mathrm{i}\to s_\mathrm{f}}x_\mathrm{i}+R_{22}^{s_\mathrm{i}\to s_\mathrm{f}}x_\mathrm{i}'+R_{26}^{s\to s_\mathrm{f}}\delta_s(z_s) \quad (1)$$

where R_{ij} are the entries of the linear transport matrix. Notice that the entries R_{11}, R_{12}, R_{21}, and R_{22} are for the transport matrix from s_i to s_f, whereas the entries R_{16} an R_{26} are for transport starting from the point s within the beamline.

The notation emphasizes the dependence of the particle energy change on the arclength coordinate s and longitudinal coordinate z_s at s, with the latter in general also depending on s. Assuming a one-to-one correspondence[1] between z_s and the particle longitudinal coordinate z_f at s_f, we can write $\delta_s(z_s)=\delta_s(z_s(z_\mathrm{f}))\equiv\delta_s(z)$, where to keep the notation simple we dropped the subscript from z_f.

Integrating over the whole dispersive region:

$$x(s_\mathrm{f})=x_\beta(s_\mathrm{f})+\int_{s_\mathrm{i}}^{s_\mathrm{f}}R_{16}^{s\to s_\mathrm{f}}\frac{d\delta_s(z)}{ds}ds\equiv x_\beta(s_\mathrm{f})+\hat{x}(z), \quad (2)$$

$$x'(s_\mathrm{f})=x_\beta{}'(s_\mathrm{f})+\int_{s_\mathrm{i}}^{s_\mathrm{f}}ds\,R_{26}^{s\to s_\mathrm{f}}\frac{d\delta_s(z)}{ds}\equiv x'_\beta(s_\mathrm{f})+\hat{x}'(z) \quad (3)$$

where $x_\beta(s_\mathrm{f})=R_{11}^{s_\mathrm{i}\to s_\mathrm{f}}x_\mathrm{i}+R_{12}^{s_\mathrm{i}\to s_\mathrm{f}}x_\mathrm{i}'$, $x'_\beta(s_\mathrm{f})=R_{21}^{s_\mathrm{i}\to s_\mathrm{f}}x_\mathrm{i}+R_{22}^{s_\mathrm{i}\to s_\mathrm{f}}x_\mathrm{i}'$, and the equalities on the RHS serve as definitions for $\hat{x}$ and $\hat{x}'$.

Assuming for simplicity that the unperturbed beam is centered $\langle x_\beta\rangle=0$, where $\langle\cdot\rangle$ is the average on the bunch population in the

[1] This requires that we neglect the longitudinal slippage due to the beam finite transverse emittance.

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6D phase space, the central 2nd-moment of the beam distribution in the horizontal coordinate at s_f reads

$$\langle (x-\langle x\rangle)^2\rangle = \langle (x_\beta - \langle x_\beta\rangle + \hat{x} - \langle \hat{x}\rangle)^2\rangle = \langle (x_\beta + \Delta x)^2\rangle = \langle x_\beta^2\rangle + \langle (\Delta x)^2\rangle \quad (4)$$

having denoted $\Delta x = \hat{x} - \langle \hat{x}\rangle$, and exploited the lack of correlation between the CSR-induced energy kick and transverse coordinate: $\langle x_\beta \Delta\hat{x}\rangle = 0$, valid in a 1D approximation. We have

$$\langle (\Delta x)^2\rangle = \langle \hat{x}^2\rangle - \langle \hat{x}\rangle^2 = \int dz\lambda \left[\int R_{16}^{s\to s_f} \frac{d\delta_s(z)}{ds} ds\right]^2 - \left[\int dz\lambda \int R_{16}^{s\to s_f} \frac{d\delta_s(z)}{ds} ds\right]^2 \quad (5)$$

where $\lambda = \lambda(z)$ is the normalized beam longitudinal density at the observation point at the exit of the beamline, with similar expressions holding for $\langle (\Delta x')^2\rangle$ and $\langle \Delta x \Delta x'\rangle$.

The rms projected (geometric) emittance is defined as the determinant of the covariance matrix

$$\boldsymbol{\sigma} = \begin{pmatrix} \langle (x+\Delta x)^2\rangle & \langle (x+\Delta x)(x'+\Delta x')\rangle \\ \langle (x+\Delta x)(x'+\Delta x')\rangle & \langle (x'+\Delta x')^2\rangle \end{pmatrix}. \quad (6)$$

Observed at the exit s_f of the beamline the emittance reads

$$\varepsilon_x^2 = \varepsilon_{x0}^2 + \varepsilon_{x0}(\beta_x \langle \Delta x'^2\rangle + 2\alpha_x \langle \Delta x \Delta x'\rangle + \gamma_x \langle \Delta x^2\rangle) + \Delta\varepsilon_x^2 \quad (7)$$

where ε_{x0} is the beam unperturbed emittance, $\alpha_x = -\langle xx'\rangle/\varepsilon_{x0}$, $\beta_x = \langle x^2\rangle/\varepsilon_{x0}$, $\gamma_x = (1+\alpha_x^2)/\beta_x$ are the Twiss functions at s_f, and

$$\Delta\varepsilon_x^2 = \langle \Delta x^2\rangle\langle \Delta x'^2\rangle - \langle \Delta x \Delta x'\rangle^2. \quad (8)$$

We now specialize the model to the case of stationary longitudinal density and 1D steady-state CSR model [12], factoring out the s dependence $d\delta_s(z)/ds \equiv g(s)\hat{\delta}'(z)$, where $g(s) = [R_0/R(s)]^{2/3}$ for s inside a bend with a radius of curvature $R(s)$, and $g(s) = 0$ outside the bends (R_0 is a constant introduced for convenience) and

$$\hat{\delta}'(z) = \frac{2mc^2 r_c N_b}{3^{1/3} R_0^{2/3}} \int_z^\infty \frac{d\lambda(z')}{dz'} \frac{dz'}{(z'-z)^{1/3}}. \quad (9)$$

In the above expression, N_b is the bunch population, m and r_c are the electron mass and the classical radius. Notice that we are adopting the convention that a particle in the bunch head has $z<0$ and $\delta'(z)>0$ denotes energy gain.

Eq. (5) then simplifies to

$$\langle (\Delta x)^2\rangle = \left[\int ds g(s) R_{16}^{s\to s_f}\right]^2 \langle (\Delta\hat{\delta}')^2\rangle \quad (10)$$

where $\langle (\Delta\hat{\delta}')^2\rangle = \langle (\hat{\delta}')^2\rangle - \langle \hat{\delta}'\rangle^2$. Similarly

$$\langle (\Delta x')^2\rangle = \left[\int ds g(s) R_{26}^{s\to s_f}\right]^2 \langle (\Delta\hat{\delta})^2\rangle \quad (11)$$

$$\langle \Delta x \Delta x'\rangle = \left[\int ds g(s) R_{16}^{s\to s_f}\right]\left[\int ds g(s) R_{26}^{s\to s_f}\right] \langle (\Delta\hat{\delta})^2\rangle. \quad (12)$$

Notice that because position and angular displacement are correlated, $\langle \Delta x \Delta x'\rangle^2 = \langle \Delta x^2\rangle\langle \Delta x'^2\rangle$, in the expression for the rms emittance (7) the last term vanishes, $\Delta\varepsilon_x^2 = 0$.

2.2. Minimization of emittance growth

Eq. (7) suggests two possible strategies for controlling emittance growth: (i) choosing $\alpha_x, \beta_x, \gamma_x$ appropriately; (ii) minimizing (ideally set to zero) the second moments $\langle \Delta x^2\rangle$, $\langle \Delta x'^2\rangle$ and $\langle \Delta x \Delta x'\rangle$. The case (i) is briefly treated in [13].

Here we discuss (ii). Emittance growth is avoided if, by choosing the optics appropriately, the following two integrals vanish ($i=1,2$):

$$\int ds g(s) R_{i6}^{s\to s_f} = 0. \quad (13)$$

These integrals vanish, in particular, if we can decompose the beamline $[s_i, s_f]$ into two subsets $S_1 \cup S_2 = [s_i, s_f]$ is such a way that for each s_1 in S_1 there exists a unique $s_2 > s_1$ in S_2 so that

$$g(s_1) R_{i6}^{s_1\to s_f} = -g(s_2) R_{i6}^{s_2\to s_f}. \quad (14)$$

Having defined the vectors

$$\boldsymbol{r}_s = \begin{pmatrix} R_{16}^{s\to s_f} \\ R_{26}^{s\to s_f} \end{pmatrix}, \quad \boldsymbol{\eta}_s = \begin{pmatrix} \eta(s) \\ \eta'(s) \end{pmatrix} \quad (15)$$

where $\eta(s)$ and $\eta'(s)$ are the dispersion function and the derivative respectively at s, it can be shown that (Appendix A) they are related by

$$\boldsymbol{r}_s = -\boldsymbol{R}_{2\times 2}^{s\to s_f} \boldsymbol{\eta}_s \quad (16)$$

where $\boldsymbol{R}_{2\times 2}^{s\to s_f}$ is the 2×2 transfer matrix for the betatron motion from s to the exit s_f of the beamline. Suppose $g(s_1) = g(s_2)$, so that we can cast (14) in the form

$$-\boldsymbol{R}_{2\times 2}^{s_1\to s_f} \boldsymbol{\eta}(s_1) = \boldsymbol{R}_{2\times 2}^{s_2\to s_f} \boldsymbol{\eta}(s_2) \quad (17)$$

and consider two subcases:

1. $\boldsymbol{\eta}_{s_2} = \boldsymbol{\eta}_{s_1}$; naturally occurring in a periodic lattice. Eq. (17) becomes
$$-\boldsymbol{R}_{2\times 2}^{s_1\to s_f} = \boldsymbol{R}_{2\times 2}^{s_2\to s_f}. \quad (18)$$
On the other hand, the matrix in the LHS of (18) can be decomposed as
$$\boldsymbol{R}_{2\times 2}^{s_1\to s_f} = \boldsymbol{R}_{2\times 2}^{s_2\to s_f} \boldsymbol{R}_{2\times 2}^{s_1\to s_2}. \quad (19)$$
Inserting (19) in (18) we find
$$\boldsymbol{R}_{2\times 2}^{s_1\to s_2} = -1 \quad (20)$$
i.e. compensation is achieved by separating CSR kicks by π in the horizontal betatron phase.
2. $\boldsymbol{\eta}_{s_2} = -\boldsymbol{\eta}_{s_1}$; as, for example, in the FERMI dog-leg [9] at the end of the Linac consisting of two double-bend achromats with antisymmetric dispersion function with respect to the dogleg mid point. Repeating the above steps, we conclude that the condition for vanishing emittance growth is now
$$\boldsymbol{R}_{2\times 2}^{s_1\to s_2} = \boldsymbol{1}. \quad (21)$$
In the FERMI transfer line consisting of a sequence of 4 dipoles, condition (21) is realized by setting a 2π phase advance from the 1st to 3rd and from the 2nd to 4th dipole.

We should point out that while these two subcases yield simple and practical recipes for designing transfer lines they do not exhaust all cases of interest. See Ref. [13] for an example of CSR compensation in a transport line that cannot be reduced to cancelation by set values of the betatron phase advances between binary kicks.

3. R-matrix vs. 'curly'-$\mathcal{H}$ formalism

Consider the simple scenario where the effect of CSR can be approximated as occurring as a single kick along the beamline localized over the short distance Δs at s as, for example, in the 4th bend of a standard bunch-compressor chicane where the beam is the shortest (and the CSR effect the largest) and approximately stationary.

We want show that the expression (7) for the emittance can be rephrased in terms of the dispersion invariant $\mathcal{H}$

$$\mathcal{H} = \eta_s^2 \gamma_x(s) + 2\alpha_x(s)\eta_s\eta_s' + \eta_s'^2 \beta_x(s). \quad (22)$$

Once again, we recognize that $\Delta\varepsilon_x^2=0$ and write the projected emittance as

$$\varepsilon_x^2 \simeq \varepsilon_{x0}^2+\varepsilon_{x0}\left[\beta_x(s_f)[R_{26}^{s\to s_f}]^2+2\alpha_x(s_f)R_{16}^{s\to s_f}R_{26}^{s\to s_f}+\gamma_x(s_f)[R_{16}^{s\to s_f}]^2\right]\sigma_{\Delta\delta}^2 \tag{23}$$

with $\sigma_{\Delta\delta}^2=[g(s)\Delta s]^2\langle(\Delta\hat{\delta}')^2\rangle$ being the CSR-induced rms relative energy spread. In writing the above equation we emphasized that the relevant Twiss functions are those at the exit of the beam line at s_f. The second term on the RHS of (23) can be written in a more compact form as

$$\varepsilon_{x0}\boldsymbol{r}_s\cdot \mathrm{T}_{s_f}^{-1}\boldsymbol{r}_s\sigma_{\Delta\delta}^2 \tag{24}$$

where $\boldsymbol{r}_s$ was defined in (15) and T is the Twiss matrix

$$\mathrm{T}=\begin{pmatrix}\beta_x & -\alpha_x\\ -\alpha_x & \gamma_x\end{pmatrix} \tag{25}$$

with inverse

$$\mathrm{T}^{-1}=\begin{pmatrix}\gamma_x & \alpha_x\\ \alpha_x & \beta_x\end{pmatrix}. \tag{26}$$

We now use the relationship (16) to write

$$\begin{aligned}\boldsymbol{r}_s\cdot \mathrm{T}_{s_f}^{-1}\boldsymbol{r}_s &= \boldsymbol{R}_{2\times2}^{s\to s_f}\boldsymbol{\eta}_s\cdot \mathrm{T}_{s_f}^{-1}\boldsymbol{R}_{2\times2}^{s\to s_f}\boldsymbol{\eta}_s\\ &= \boldsymbol{\eta}_s\cdot[\boldsymbol{R}_{2\times2}^{s\to s_f}]^T\mathrm{T}_{s_f}^{-1}\boldsymbol{R}_{2\times2}^{s\to s_f}\boldsymbol{\eta}_s\end{aligned} \tag{27}$$

where we have made use of the properties of the scalar product to move $\boldsymbol{R}_{2\times2}^{s\to s_f}$ from the left to the right of '$\cdot$'. Next, it is a matter of recognizing that T propagates like the beam co-variance matrix

$$\mathrm{T}_{s_f}=\boldsymbol{R}_{2\times2}^{s\to s_f}\mathrm{T}_s[\boldsymbol{R}_{2\times2}^{s\to s_f}]^T \tag{28}$$

or

$$\mathrm{T}_{s_f}^{-1}=[\boldsymbol{R}_{2\times2}^{s\to s_f}]^{-T}\mathrm{T}_s^{-1}[\boldsymbol{R}_{2\times2}^{s\to s_f}]^{-1} \tag{29}$$

to write (27) as

$$\boldsymbol{r}_s\cdot \mathrm{T}_{s_f}^{-1}\boldsymbol{r}_s=\boldsymbol{\eta}_s\cdot \mathrm{T}_s^{-1}\boldsymbol{\eta}_s=\eta_s^2\gamma_x(s)+2\alpha_x(s)\eta_s\eta_s'+\eta_s'^2\beta_x(s). \tag{30}$$

In conclusion, the beam emittance at the exit of the dispersive region is equivalent to

$$\varepsilon_x^2=\varepsilon_{x0}^2+\varepsilon_{x0}\mathcal{H}(s)\sigma_{\Delta\delta}^2. \tag{31}$$

It is worth noticing that in contrast to Eq. (23), where the Twiss functions are relative to the observation point s_f at the exit of the beamline, in (31) the Twiss functions entering into the expression for the dispersion invariant $\mathcal{H}$ are at the position s of the CSR kick.

In more general cases where CSR is not highly concentrated the expression of the emittance growth in terms of the dispersion invariant is less accurate and Eq. (23) should be used instead. We stress that one should not be tempted to integrate the RHS of (31) over s as a way to get an accurate estimate of CSR emittance growth contributed by the whole beamline of interest.

4. Conclusions

We have reviewed the linear R-matrix formalism describing CSR-induced emittance growth in transport lines and recovered lattice design prescriptions yielding vanishing emittance growth for lattice with periodic or antisymmetric dispersion functions. Finally, we established the connection between a description of the emittance growth in terms of the R-matrix and the one in terms of the dispersion invariant.

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Appendix A

The goal in this appendix is to prove Eq. (16). The most general linear matrix for transport trough a sequence of bends, quads, and drifts has the form

$$\boldsymbol{R}=\begin{pmatrix}R_{11} & R_{12} & 0 & R_{16}\\ R_{21} & R_{22} & 0 & R_{26}\\ R_{51} & R_{52} & 1 & R_{56}\\ 0 & 0 & 0 & 1\end{pmatrix}. \tag{A.1}$$

Incidentally, notice that symplecticity implies

$$\begin{pmatrix}R_{16}\\ R_{26}\end{pmatrix}=-\begin{pmatrix}R_{11} & R_{12}\\ R_{21} & R_{22}\end{pmatrix}\begin{pmatrix}R_{52}\\ -R_{51}\end{pmatrix}=-\boldsymbol{R}_{2\times2}\begin{pmatrix}R_{52}\\ -R_{51}\end{pmatrix} \tag{A.2}$$

where the second equality defines $\boldsymbol{R}_{2\times2}$. The inverse reads

$$\boldsymbol{R}^{-1}=\begin{pmatrix}R_{22} & -R_{12} & 0 & R_{52}\\ -R_{21} & R_{11} & 0 & -R_{51}\\ -R_{26} & R_{16} & 1 & -R_{56}\\ 0 & 0 & 0 & 1\end{pmatrix}. \tag{A.3}$$

If the beamline $[s_i, s_f]$ is an achromat, by definition the entries $R_{16}^{s_i\to s}\equiv\eta(s)$ and $R_{26}^{s_i\to s}\equiv\eta'(s)$ where $s\in[s_i, s_f]$, identify the dispersion function and its derivative. We are interested in finding an expression for the entries $R_{16}^{s\to s}$, and $R_{26}^{s\to s}$ in terms of $\eta(s)$ and $\eta'(s)$.

As the longitudinal dynamics is inconsequential in this calculation, we rewrite the transfer matrix $\boldsymbol{R}$ from s_i to s as a 3×3 matrix

$$\boldsymbol{R}^{s_i\to s}=\begin{pmatrix}\boldsymbol{R}_{2\times2}^{s_i\to s} & \boldsymbol{\eta}_s\\ 0 & 1\end{pmatrix} \tag{A.4}$$

where $\boldsymbol{\eta}_s=(\eta(s),\eta'(s))$ is the vector defined in (15). The inverse of this matrix $\boldsymbol{R}^{s_i\to s}$ is [see Eqs. (A.2) and (A.3)]:

$$[\boldsymbol{R}^{s_i\to s}]^{-1}=\begin{pmatrix}[\boldsymbol{R}_{2\times2}^{s_i\to s}]^{-1} & -[\boldsymbol{R}_{2\times2}^{s_i\to s}]^{-1}\boldsymbol{\eta}_s\\ 0 & 1\end{pmatrix}. \tag{A.5}$$

Next, we write the matrix from s to s_f; using the definition of $\boldsymbol{r}_s$ from Eq. (15) we have

$$\boldsymbol{R}^{s\to s_f}=\begin{pmatrix}\boldsymbol{R}_{2\times2}^{s\to s_f} & \boldsymbol{r}_s\\ 0 & 1\end{pmatrix}. \tag{A.6}$$

Combining the two matrices from s_i to s and from s to s_f we have $\boldsymbol{R}^{s_i\to s_f}=\boldsymbol{R}^{s\to s_f}\boldsymbol{R}^{s_i\to s}$ and therefore

$$\begin{aligned}\boldsymbol{R}^{s\to s_f} &= \boldsymbol{R}^{s_i\to s_f}[\boldsymbol{R}^{s_i\to s}]^{-1}\\ &= \begin{pmatrix}\boldsymbol{R}_{2\times2}^{s_i\to s_f} & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}[\boldsymbol{R}_{2\times2}^{s_i\to s}]^{-1} & -[\boldsymbol{R}_{2\times2}^{s_i\to s}]^{-1}\boldsymbol{\eta}_s\\ 0 & 1\end{pmatrix}\\ &= \begin{pmatrix}\boldsymbol{R}_{2\times2}^{s_i\to s_f}[\boldsymbol{R}_{2\times2}^{s_i\to s}]^{-1} & -\boldsymbol{R}_{2\times2}^{s_i\to s_f}[\boldsymbol{R}_{2\times2}^{s_i\to s}]^{-1}\boldsymbol{\eta}_s\\ 0 & 1\end{pmatrix}.\end{aligned} \tag{A.7}$$

Comparing the two RHSs of Eqs. (A.6) and (A.7) we conclude

$$\begin{aligned}\boldsymbol{r}(s) &= -\boldsymbol{R}_{2\times2}^{s_i\to s_f}[\boldsymbol{R}_{2\times2}^{s_i\to s}]^{-1}\boldsymbol{\eta}_s\\ &= -\boldsymbol{R}_{2\times2}^{s\to s_f}\boldsymbol{R}_{2\times2}^{s_i\to s}[\boldsymbol{R}_{2\times2}^{s_i\to s}]^{-1}\boldsymbol{\eta}_s\\ &= -\boldsymbol{R}_{2\times2}^{s\to s_f}\boldsymbol{\eta}_s,\end{aligned} \tag{A.8}$$

the same as Eq. (16).

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